

DAY 3

TOPIC

- RAM and StorageTheory
- Types of RAM (DDR3, DDR4, DDR5)
- storage (HDD vs. SSD, NVMe)
- Remove and reinstall RAM and storage drives.

TASKS:1

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

RESULT

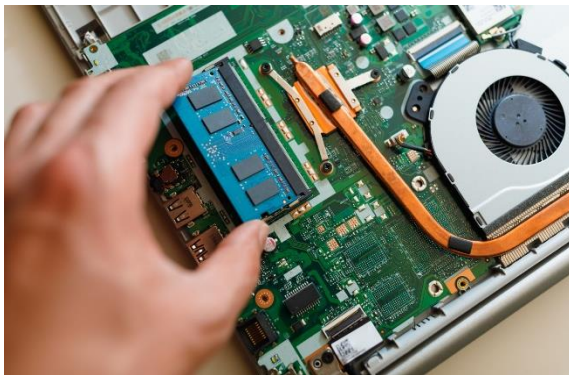
FEATURE	DDR3	DDR4	DDR5
Typical speed	800-2133MT/s	1600-3200 MT/s	4800-8400+MT/s
Voltage	1.5V(1.35V for DDR3L)	1.2V	1.1V
Bandwidth	Lower	higher	Much higher
Typical use	Older PCs and laptop	Many older/current PCs	Modern and high performance PCs
Availability	Mostly obsolete	Still comman	Current standard
Example use	Legacy desktops	Office,gaming	Gaming,workstation

Summary: DDR5 offers the highest speed and bandwidth while using the lowest standard voltage. DDR3 is the oldest generation, while DDR4 sits between DDR3 and DDR5 in performance.

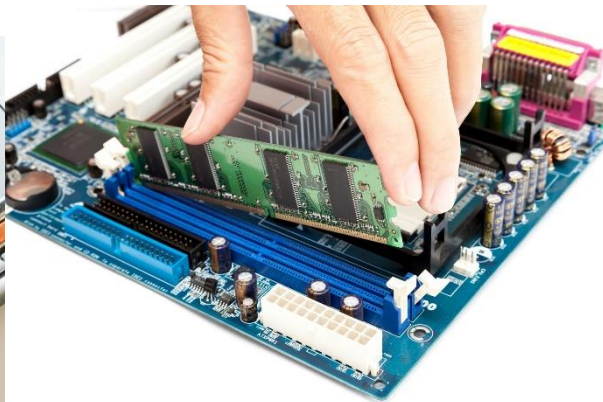
TASKS:2

Open your laptop or desktop, locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

RESULT



Install



Reinstall

PRATICAL METHOD

Tools & Materials Required:

- Laptop or Desktop PC
- Compatible RAM module
- Anti-static wrist strap (optional/recommended)

- Screwdriver set (if opening laptop back panel)

Procedure / Steps:

1. Safety Preparation:

- Power off the computer completely.
- Disconnect the main power cable and remove the battery (if accessible).
- Press and hold the power button for 5 seconds to drain any residual electricity.

2. Locating and Removing RAM:

- Open the computer side panel (desktop) or bottom cover (laptop).
- Locate the RAM slots on the motherboard.
- Unclip the side retaining latches to release the current RAM module and gently pull it out.

3. Reinstalling the RAM:

- Align the notch on the RAM stick's gold contacts with the ridge in the motherboard slot.
- For Laptop (SO-DIMM): Insert the RAM at a 30-degree angle, then push it downward until the side clips snap into place.

- For Desktop (DIMM): Position the RAM vertically above the slot and push down firmly until the lock latches snap shut automatically.

4. Verification & Testing:

- Reattach the outer cover/casing and reconnect the power cable.
- Turn on the system and verify that it boots into the Operating System normally.
- Open *Task Manager* (Windows) or *About This Mac* to confirm the total RAM size is correctly detected.

TASKS:3

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case that would benefit most from that storage type.

RESULT

HDD vs SSD vs NVMe

Storage type	key difference	speed	Best use case/app
HDD	Uses spinning magnetic disks: cheaper and offers large capacity	Slow	GOOGLE DRIVES: Storing large files and backup
SSD	Uses flash memory with no moving parts: faster and quieter than HDD	Fast	MICROSOFT WORD:quick startup and opening document
NVme SSD	Uses the PCIe interface,providing much higher data-transfer speed than sata SSDs	Very fast	ADOBE PHOTOSHOP: faster loading and handling of large files

Three key differences:

- 1. **Technology:** HDD uses magnetic disks, SSD uses flash memory, and NVMe uses PCIe-based flash storage.
- 2. **Performance:** HDD is slowest, SSD is faster, and NVMe is generally the fastest.

3. **Use:** HDD is best for inexpensive bulk storage, SSD for everyday computing, and NVMe for demanding applications.

TASKS:4

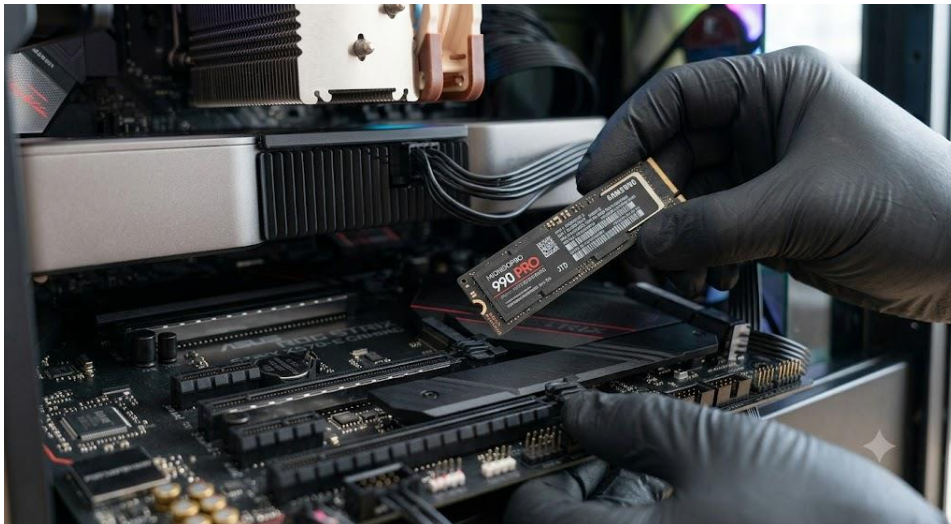
Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

RESULT

REMOVAL PROCESS

- ❑ First, **shut down the computer completely.**
- ❑ **Disconnect the power cable** from the computer.
- ❑ Open the CPU cabinet side panel carefully.
- ❑ Locate the **SSD** inside the cabinet.
- ❑ Identify the **SATA data cable** and **SATA power cable** connected to the SSD.
- ❑ Carefully disconnect both cables from the SSD.

- ❓ Remove the screws or bracket holding the SSD in place.
- ❓ Gently remove the SSD from the drive bay.
- ❓ Do not touch the **gold contacts or connectors** unnecessarily.
- ❓ Take a clear photo of the SSD and its **SATA connector** for the assignment





REINSTALLATION PROCESS

Place the SSD back into its original position.

Secure it with the screws/bracket.

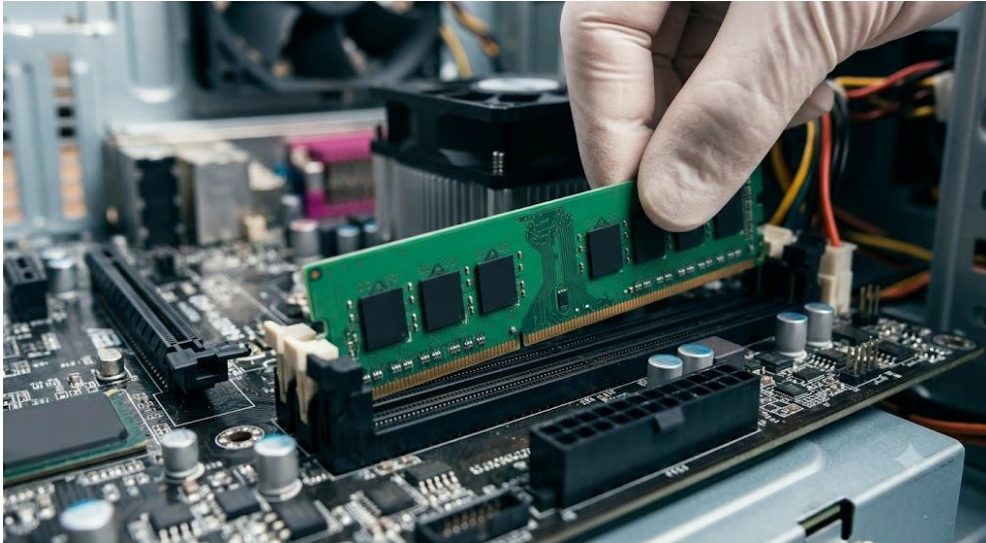
Connect the SATA data cable.

Connect the SATA power cable.

Close the computer cabinet.

Reconnect the power cable and start the computer.

Open This PC or Disk Management and check that the SSD is detected.



CONCLUSION

- ➡ The SSD was successfully removed and reinstalled. After restarting the computer, the SSD was recognized successfully by the system.”